

HYPERTEXT MARKUP LANGUAGE

Multimedia Elements:

The Power of Media Elements in HTML

BEFORE
(Without Media Elements)

AFTER
(With Media Elements)

Same Content, Better Experience!

THE OCEAN

The ocean covers more than 70% of Earth's surface.

It is home to millions of species of plants and animals.

The ocean plays a very important role in our climate and environment.

We must protect the ocean for a better future.

Home to Life
It is home to millions of species of plants and animals.

Climate Guardian
The ocean plays a very important role in our climate and environment.

Protect Our Future
We must protect the ocean for a better future.

- Looks plain and boring
- Hard to engage the reader
- No visualization or emotion
- Information feels dry and heavy

- Visually appealing and engaging
- Easy to understand and remember
- Creates emotion and connection
- Information becomes powerful and memorable

The Difference Audio Makes in HTML

BEFORE
(Without Audio)

AFTER
(With Audio)

Nature and Its Sounds

The forest is full of amazing sounds. You can hear birds chirping, leaves rustling, and the river flowing. These sounds create a peaceful and relaxing environment.

You can only read about the sounds. You can't hear them.

Now you can listen and feel the sounds bringing the content to life!

- Information is only in text.
- Harder to imagine the experience.
- Less engaging and memorable.

- Adds real sounds to the content.
- Helps users imagine and connect.
- More engaging and memorable experience.

The Difference Video Makes in HTML

BEFORE
(Without Video)

AFTER
(With Video)

How a Solar Panel Works

Solar panels convert sunlight into electricity through photovoltaic cells. When sunlight hits the cells, it excites electrons and creates electric current.

This electricity can be used to power homes, schools, and many devices.

The Experience

- Needs more imagination
- Harder to understand the full process
- Less engaging and easy to skim
- Information feels flat and dry

You read about it, but it's hard to see it in action.

How a Solar Panel Works

0:00 / 1:18

The Experience

- See the process in action
- Easier to understand and remember
- More engaging and enjoyable
- Information feels real and powerful

You see it, understand it, and remember it better.

VS

Video turns information into an experience. It connects, explains, and leaves a lasting impact.

HYPERTEXT MARKUP LANGUAGE

HTML Multimedia

HTML multimedia refers to different types of media content that can be displayed or played on a webpage. Multimedia makes webpages more interactive, informative, and visually appealing by combining text with images, sound, videos, and external media sources.

Common multimedia elements include:

- Images
- Image Maps
- Audio
- Video

Why Use Multimedia?

Multimedia helps to:

- Improve user engagement
- Explain concepts visually
- Add entertainment value
- Present information more effectively
- Create interactive web experiences
- Enhance website design

HTML Images

Images are visual elements displayed on a webpage. HTML uses the `<img>` tag to insert images.

The `<img>` tag is an empty element, meaning it does not require a closing tag.

Basic Syntax

```

```

Main Attributes of Images:

Attribute	Purpose
Src	Specifies image location
Alt	Alternative text if image cannot load
Width	Sets image width
Height	Sets image height
Title	Displays tooltip text

Example

```

```

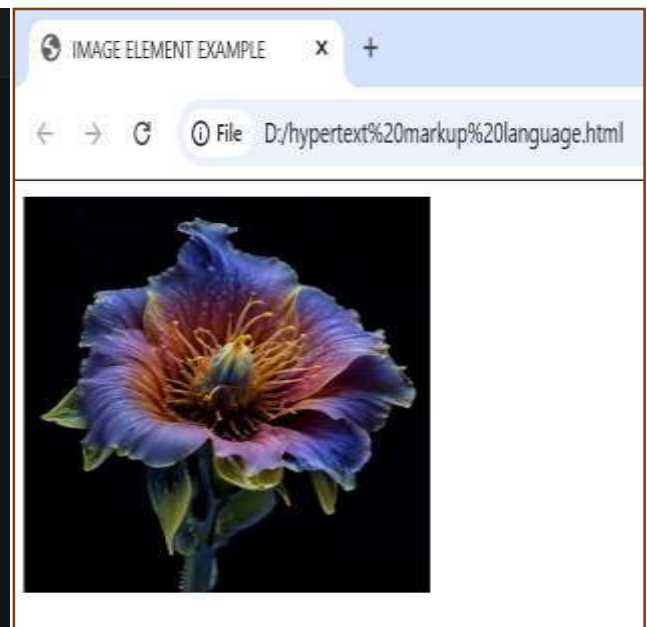
Setting Image Size

```

```




```
hypertext markup language.html X
D: > <> hypertext markup language.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>IMAGE ELEMENT EXAMPLE</title>
5  </head>
6  <body>
7  |   
11 </body>
12 </html>
13
```



Why Use Alt Text?

The alt attribute:

- Helps visually impaired users
- Appears when images fail to load
- Improvessibility
- Helps search engines understand images

Example:

```

```

Image Maps

An image map divides a single image into multiple clickable regions. Different parts of one image can perform different actions.

Real-Life Example

Imagine a world map:

- ❑ Clicking India opens India's page.
- ❑ Clicking Japan opens Japan's page.
- ❑ Clicking USA opens USA's page.

One image performs multiple navigation functions.

Image Map Structure

Image maps use:

- ❑
- ❑ <map>
- ❑ <area>



Example

```

<map name="worldmap">
<area shape="rect" coords="20,20,120,120" href="india.html">
</map>
```

Important Image Map Attributes

	Attribute	Purpose
Shape	→	Defines clickable shape
Cords	→	Defines coordinates
Href	→	Destination link
Alt	→	Alternative text

Image Map Shapes

Rectangle

shape="rect"

Creates a rectangular clickable area.

Circle

shape="circle"

Creates a circular clickable area.

Polygon

shape="poly"

Creates a custom-shaped clickable area.

HTML Audio

Audio allows webpages to play sound files directly inside browsers.

The <audio> element is used for audio playback

Why Use Audio?

Audio is useful for:

- Music players
- Podcasts
- Voice instructions
- Learning platforms
- Notifications

Basic Syntax

```
<audio controls>
  <source src="music.mp3">
</audio>
```



Audio Controls

The controls attribute displays:

- Play button
- Pause button
- Volume control
- Progress bar

Example

```
<audio controls>
```

```
  <source src="song.mp3" type="audio/mpeg">
```

```
</audio>
```



Audio Attributes:

Attribute	Purpose
Controls	Shows player controls
Autoplay	Starts automatically
Loop	Repeats audio
Muted	Starts muted
Preload	Loads audio before playback

Autoplay Example

```
<audio autoplay controls>
```

```
  <source src="song.mp3">
```

```
</audio>
```

Loop Example

```
<audio loop controls>
```

```
  <source src="song.mp3">
```

```
</audio>
```



Video Attributes:

Attribute	Purpose
controls	Displays video controls
Width	Sets video width
Height	Sets video height
Autoplay	Plays automatically
Loop	Repeats video
Muted	Starts without sound
Poster	Displays preview image

Poster Image

A poster image appears before the video starts.

```
<video controls  
  poster="thumbnail.jpg">  
  
  <source src="movie.mp4">  
  
</video>
```

Autoplay Video

```
<video autoplay muted>  
  
  <source src="video.mp4">  
  
</video>
```

Most browsers require muted autoplay.

Multiple Video Formats

```
<video controls>  
  
  <source src="movie.mp4"  
    type="video/mp4">  
  
  <source src="movie.webm"  
    type="video/webm">
```



<https://elevatme.netlify.app/>



<https://yaganti-spandana.netlify.app/>